```
for (int i = 0; i < 3; i++) {
    printf("%d ", *(ptr + i));  // Accessing array elements using pointer arithmetic
}
```

Accessing Array Elements Using Pointers

You can access array elements through pointers by using indexing (`ptr[i]`) or dereferencing (`*(ptr + i)`):

```
printf("%d", ptr[1]);  // Same as *(ptr + 1)
```

Common Pitfalls and Best Practices

1. **Memory Leaks:** Ensure every `malloc` or `calloc` is paired with a `free` .
2. **Dangling Pointers:** Set pointers to `NULL` after freeing memory.
3. **Buffer Overflows:** Avoid accessing memory beyond allocated limits.
4. **NULL Checks:** Always check if memory allocation was successful.

Conclusion

Pointers and dynamic memory management are powerful tools that enable efficient handling of memory